

Pre-Board Examination - 2079

Class : XII

Subject: Physics (1021)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

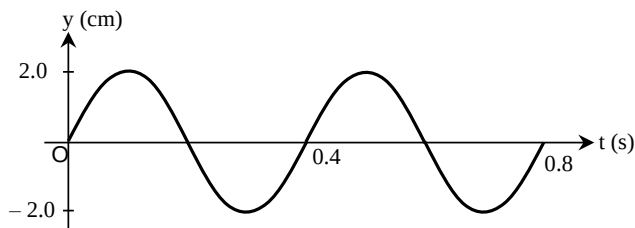
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

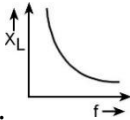
Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

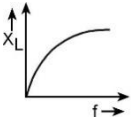
1. A particle executing SHM has total energy of 5J. Then its maximum potential energy is
 a. 5J b. 10J c. 2.5J d. -10J
2. The displacement versus time graph of a particle executing SHM is as shown in the figure. The maximum velocity of particle is
 a. 31.4 cm/s
 b. 3.14 cm/s
 c. 0.2 cm/s
 d. 1 cm/s

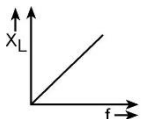


3. Viscosity of liquid and gas, with the increase in temperature
 a. Decreases in liquid and increases in gas
 b. Increases in liquid and decreases in gas
 c. Increases in both
 d. Decreases in both
4. The coefficient of performance of a refrigerator working between ice point and room temperature 27°C is nearly
 a. 10 b. 15 c. 20 d. 25

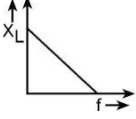
5. Internal energy of a real gas depends on
 a. Volume and temperature b. Pressure and temperature
 c. Volume only d. Temperature only
6. Which one of the following directly affects the quality of sound?
 a. Shape of the source b. Frequency
 c. Intensity d. Wave form
7. A diffraction pattern is observed using red light. Which of the following is observed if red light is replaced by blue light?
 a. There is no change in width of diffraction bands
 b. Diffraction bands become narrow and crowded
 c. Diffraction bands become broader and farther
 d. Bands disappear
8. If 2% of main current is passed through galvanometer of resistance 49Ω . The shunt resistance used is
 a. 0.1Ω b. 1Ω c. 0.2Ω d. 2Ω
9. The material suitable for making electromagnet should have
 a. high retentivity and high coercivity
 b. low retentivity and low coercivity
 c. high retentivity and low coercivity
 d. low retentivity and low coercivity
10. Which of the following figure represents variation of inductive reactance with frequency?
- 

a.



b.
- 

c.



d.
11. Cut-off wavelength of X-rays coming from a Coolidge tube depends on
 a. target material
 b. accelerating voltage
 c. temperature of the filament
 d. separation between temperature and filament

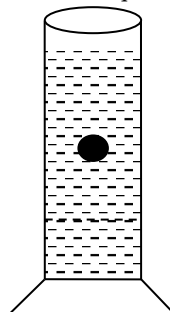
Group - B

Give short answer to the following questions.

[8 × 5 = 40]

1. Figure alongside shows a steel ball falling freely through a viscous liquid.

- Sketch the various forces acting on the ball. [1]
- Under what condition ball attains the terminal velocity? Sketch the velocity-time graph for the motion of ball. [2]
- Derive the expression for the coefficient of viscosity of liquid for the set up shown in figure. [2]



OR

The liquid molecules on free surface of liquid possess potential energy called surface energy as they need to overcome the downward force of attraction.

- Show that surface energy density of liquid is numerically equal to its surface tension. [2]
 - Small liquid drops are spherical while large liquid drops are oval. Why? [1]
 - Calculate the force required to pull out the flat circular plate of radius 4 cm from the surface of water neglecting its weight. The surface tension of water is $7.2 \times 10^{-3} \text{ N/m}$. [2]
2. In the absence of external torque, the total angular momentum of system is conserved.
- Explain why the earth will slow down slightly if it is struck by meteorites? [1]
 - A horizontal disc rotating freely about a vertical axis at 100 rpm. A small piece of wax of mass 10g falls vertically onto the disc and adheres to it at a distance of 9cm from the axis and the number of revolutions per minute is thereby reduced to 90. Calculate the moment of inertia of disc. [3]
 - Two spheres of equal mass but one solid and another hollow. Which one will have greater moment of inertia? Explain. [1]

3. a) First law of thermodynamics obeys principle of conservation of energy. Justify it. [1]
- b) Derive Mayer's relation. [3]
- c) In adiabatic process, no heat is exchanged but work is done, does it mean that adiabatic process violates first law of thermodynamics? [1]

OR

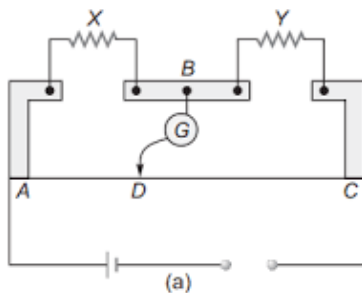
Refrigerator is a heat pump that removes the heat from lower temperature body to higher temperature surroundings.

- a) Does it violate the principle of second law of thermodynamics? [1]
- b) Show the schematic diagram of working of refrigerator. [2]
- c) At what condition, the coefficient of performance is infinity. Does it realize in practice? [2]
4. a) Distinguish between interference and diffraction. [2]
- b) "Light can be added to light to produce darkness". Justify this statement. [1]
- c) Two narrow slits 3×10^{-4} m apart are illuminated by monochromatic light of wavelength 5.9×10^{-7} m. Fringes are obtained on screen placed 0.3m behind two slits. Calculate
- i) width of fringes. [1]
- ii) width of fringes when the apparatus is immersed in liquid of refractive index 1.5 without disturbing the arrangement. [1]

OR

- a) State the Huygens principle for the propagation of light wave in medium. Using this principle verify the laws of refraction of light. [1+3]
- b) What is the drawback of Huygens wave theory? [1]

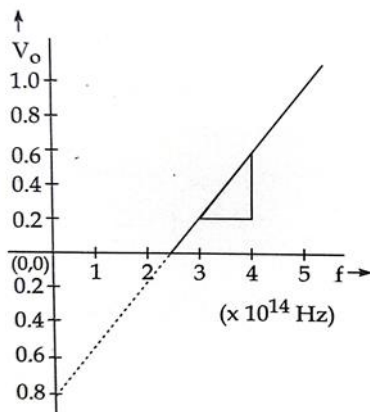
5. a) State Kirchhoff's laws of electrical circuit. [2]
- b) In meter bridge, the null point is found at a distance of 60 cm from A. If now a resistor of 5Ω is connected in series with Y, the null point occurs at 50 cm. Determine the values of X and Y. [3]



6. a) Two particles A and B of masses m and $2m$ have charges q and $2q$ respectively. Both these particles moving with velocities v_1 and v_2 respectively in same direction enter the same magnetic field B acting normally to their direction of motion. If the two forces F_A and F_B acting on them are in the ratio 1:2, find the ratio of their velocities. [2]
- b) A rectangular coil of sides ' l ' and ' b ' carrying a current ' I ' is subjected to a uniform magnetic field B . Obtain an expression for the torque acting on it. [3]

7. In the graph aside,

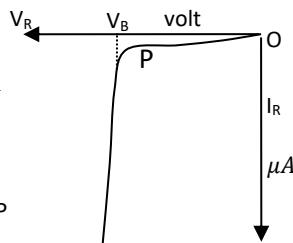
- a) What does the slope and intercept of graph represent? [2]
- b) Find Planck's constant. [2]
- c) Calculate the work function of a metal analyzing graph. [1]



8. Study the given graph and answer the questions.

- a) The characteristics of which type of biasing of P-N junction diode is shown in the figure? [1]

- b) Draw a circuit showing such type of biasing of P-N junction diode. [1]



- c) What does V_B indicate in the graph? [1]
- d) Why there is very small current (represented by OP in the graph) flowing through the diode before V_B ? What is this current called? [1]
- e) Why current rises abruptly through the diode at V_B ? [1]

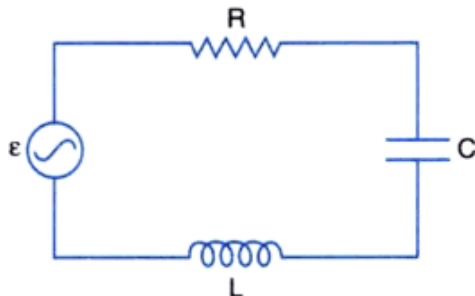
Group - C

Give long answer to the following questions.

[3 × 8 = 24]

9. a) Discuss the Newton's formula for the velocity of sound in air with Laplace correction. [3]
- b) An observer is moving towards the stationary source. If the observed pitch is 30% greater than the actual pitch, calculate the velocity of observer. (Velocity of sound in air = 330m/s). [2]
- c) Solid can support both longitudinal and transverse wave but gas can only support only longitudinal wave, why? [1]
- d) What will happen to the frequency of note emitted by organ pipe if
- temperature of air column is increased. [1]
 - diameter of pipe is increased. [1]

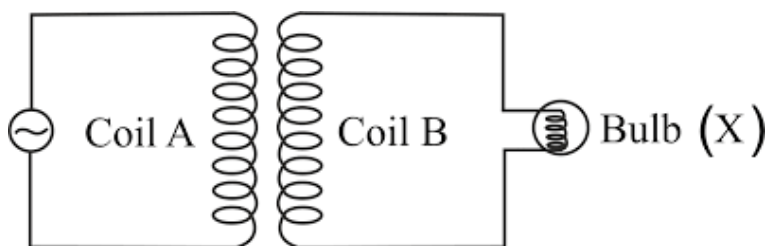
10. a) Define rms value of an ac. [1]
- b) A series LCR circuit is connected to an ac source of emf E as shown in figure.



- i) Draw phasor diagram to obtain impedance of the circuit and phase angle between the current and emf. [3]
- ii) If $L = 5\text{H}$, $C = 8\mu\text{F}$ and $R = 40\ \Omega$, determine the source frequency which drives the current in resonance. [2]
- c) A choke coil is preferred to resistor in an ac circuit. Why? [2]

OR

- a) State Lenz law. [1]
- b) The given figure shows an arrangement by which current flows through the bulb X connected with coil B, when ac is passed through the coil A.



- i) Name the principle in it. [1]
- ii) What happens in the brightness of the bulb if the number of turns of the coil B is increased? [1]
- iii) Does the bulb glow if ac source is replaced by dc source? Explain. [1]
- iv) The bulb gets dimmer if the coil B is moved upward. Why? [1]

- c) A plane circular coil has 200 turns and its radius is 0.10 m. It is connected to a battery. After switching on the circuit, a current of 2A is set up in the coil. Calculate the energy stored in the coil.

$$(\mu_o = 4\pi \times 10^{-7} \text{ Hm}^{-1}) \quad [3]$$

11. In 1913, Sir William Bragg and Sir Lawrence Bragg worked out a mathematical relation to determine interatomic distance from X-rays diffraction pattern which is called Bragg's law.

- a) Derive Bragg's law and explain how this law is used to determine crystal plane spacing. [3]
- b) An X-ray tube works at a dc p.d of 50KV. Only 0.4% of the energy of the cathode rays is converted into X-rays and heat is generated in target at the rate of 600watt. Estimate
- i) the current passed into the tube [2]
 - ii) velocity of electrons striking the target. [1]
- c) Define excitation energy and excitation potential. [2]

OR

- a) Electron is deviated in electric and magnetic fields.
- i) What path does the electron follow in electric field? Derive the equation for its trajectory. [1+2]
 - ii) An electron passes through a space without deviation. Does it mean, there is no field? [1]
- b) Define three units of activity. [2]
- c) What are p-waves and s-waves? [2]

☞ - The End - ☞